Laboratori 3: - Learning-SQL IBD

Júlia Roca Lladós
Izan Flores Tortosa
Jordi Vivas Sakuta

1

```
MariaDB [store]> create database school;
Query OK, 1 row affected (0.001 sec)
```

2

```
MariaDB [school]> create table students
    -> (student_id INT (20) NOT NULL AUTO_INCREMENT,
    -> fname VARCHAR(30) NOT NULL DEFAULT 'NO NAME PROVIDED',
    -> PRIMARY KEY (student_id)
    -> );
Query OK, 0 rows affected (0.018 sec)
```

3

```
MariaDB [school]>  create table papers
    -> ( title VARCHAR(30) NOT NULL DEFAULT 'NO TABLE PROVIDED',
    -> GRADE INT(2) NOT NULL DEFAULT '0',
    -> students_id INT (20) NOT NULL AUTO_INCREMENT,
    -> FOREIGN KEY(students_id) REFERENCES students(student_id)
    -> );
Query OK, 0 rows affected (0.038 sec)
```

4

```
MariaDB [school]>  INSERT INTO students (fname)
    ->  VALUES ("Caleb"),
    -> ("Samantha"),
    -> ("Raj"),
    -> ("Carlos"),
    -> ("Lisa");
Query OK, 5 rows affected (0.005 sec)
Records: 5  Duplicates: 0  Warnings: 0

MariaDB [school]> select * from students;
+------------+----------+
| student_id | fname    |
+------------+----------+
|          1 | Caleb    |
|          2 | Samantha |
|          3 | Raj      |
|          4 | Carlos   |
|          5 | Lisa     |
+------------+----------+
5 rows in set (0.000 sec)
```

5

```
MariaDB [school]>  INSERT INTO papers (title, GRADE, students_id)
    -> VALUES ("My First Book Report", 60, 1),
    -> ("My Second Book Report", 75, 1),
    -> ("Russian Lit Through The Ages", 94, 2),
    -> ("De Montaigne and The Art of The Essay", 98, 2),
    -> ("Borges and Magical Realism", 89, 4);
Query OK, 5 rows affected, 1 warning (0.006 sec)
Records: 5  Duplicates: 0  Warnings: 1

MariaDB [school]> select * from papers;
+--------------------------------+-------+-------------+
| title                          | GRADE | students_id |
+--------------------------------+-------+-------------+
| My First Book Report           |    60 |           1 |
| My Second Book Report          |    75 |           1 |
| Russian Lit Through The Ages   |    94 |           2 |
| De Montaigne and The Art of Th |    98 |           2 |
| Borges and Magical Realism     |    89 |           4 |
+--------------------------------+-------+-------------+
5 rows in set (0.000 sec)
```

6

```
MariaDB [school]> SELECT fname, title, GRADE
    -> FROM students, papers
    -> WHERE students.student_id = papers.students_id
    -> ORDER BY GRADE DESC;
+----------+--------------------------------+-------+
| fname    | title                          | GRADE |
+----------+--------------------------------+-------+
| Samantha | De Montaigne and The Art of Th |    98 |
| Samantha | Russian Lit Through The Ages   |    94 |
| Carlos   | Borges and Magical Realism     |    89 |
| Caleb    | My Second Book Report          |    75 |
| Caleb    | My First Book Report           |    60 |
+----------+--------------------------------+-------+
5 rows in set (0.001 sec)
```

7

```
MariaDB [school]> SELECT * FROM students
    -> LEFT JOIN papers
    -> ON students.student_id = papers.students_id;
+------------+----------+--------------------------------+-------+-------------+
| student_id | fname    | title                          | GRADE | students_id |
+------------+----------+--------------------------------+-------+-------------+
|          1 | Caleb    | My First Book Report           |    60 |           1 |
|          1 | Caleb    | My Second Book Report          |    75 |           1 |
|          2 | Samantha | Russian Lit Through The Ages   |    94 |           2 |
|          2 | Samantha | De Montaigne and The Art of Th |    98 |           2 |
|          4 | Carlos   | Borges and Magical Realism     |    89 |           4 |
|          3 | Raj      | NULL                           |  NULL |        NULL |
|          5 | Lisa     | NULL                           |  NULL |        NULL |
+------------+----------+--------------------------------+-------+-------------+
7 rows in set (0.001 sec)
```

8

```
MariaDB [school]>      SELECT
    ->      fname,
    ->      IFNULL(AVG(GRADE), 0) AS Average_Grade,
    ->      IF (IFNULL(AVG(GRADE), 0) >75,'passing','failed') AS status
    ->      FROM students
    ->      left join papers
    ->      on students.student_id = papers.students_id
    ->      group by students.fname
    ->      ORDER BY Average_Grade DESC;
+----------+---------------+---------+
| fname    | Average_Grade | status  |
+----------+---------------+---------+
| Samantha |       96.0000 | passing |
| Carlos   |       89.0000 | passing |
| Caleb    |       67.5000 | failed  |
| Raj      |        0.0000 | failed  |
| Lisa     |        0.0000 | failed  |
+----------+---------------+---------+
5 rows in set (0.001 sec)
```